



**JOMO KENYATTA UNIVERSITY
OF
AGRICULTURE & TECHNOLOGY**

SCHOOL OF OPEN, DISTANCE AND eLEARNING

P.O. Box 62000, 00200

Nairobi, Kenya

E-mail: elearning@jkuat.ac.ke

SMA 2101: Calculus I

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This presentation is intended to be covered within one week. The notes, examples and exercises should be supplemented with a good textbook. Most of the exercises have solutions/answers appearing elsewhere and accessible by clicking the green **Exercise** tag. To move back to the same page click the same tag appearing at the end of the solution/answer.

Errors and omissions in these notes are entirely the responsibility of the author who should only be contacted through the **Department of Curricula & Delivery (SODeL)** and suggested corrections may be e-mailed to elarning@jkuat.ac.ke.



LESSON 5

Implicit Functions

Explicit functions of x are functions such as $y = x^2 - 5x + 4$

In this case y is given an expression in x . If however y is given implicitly

By an equation such as $x = y^4 - y + 1$, we cannot express y in terms of x .

Consider $x = y^2$, $y = \sqrt{x} = x^{1/2}$

$\frac{dy}{dx} = \frac{1}{2}x^{-1/2} = \frac{1}{2x^{1/2}}$ or $\frac{dy}{dx} = \frac{1}{2y}$ which implies that $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$

When it is impractical to express either variable explicitly in terms of the other, we can

still differentiate both sides with respect to x .

A term like y^n can be differentiated by first differentiating



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w.r.t y then as the chain

rule demands, multiply by $\frac{dy}{dx}$, thus

$$\frac{d}{dx} (y^n) = \frac{d}{dy} (y^n) \frac{dy}{dx} = ny^{n-1} \frac{dy}{dx}$$

Similarly if we have a term of the function $x^m y^n$ then we use the product rule and obtain

$$\frac{d}{dx} (x^m y^n) = y^n \frac{d}{dx} (x^m) + x^m \frac{d}{dy} (y^n) \frac{dy}{dx} = mx^{m-1} y^n + nx^m y^{n-1} \frac{dy}{dx}$$

Example . Find $\frac{dy}{dx}$ given that $x^2 + 2xy^3 = 6$

Solution: $x^2 + 2xy^3 = 6$, differentiating both sides with respect to x , we have

$2x + 2(y^3 + x3y^2 \frac{dy}{dx}) = 0 \Rightarrow 2x + 2y^3 + 6xy^2 y' = 0$, make y' the subject

$$6xy^2 y' = -2x - 2y^3 \Rightarrow 3xy^2 y' = -x - y^3$$



$$y' = \frac{-(x+y^3)}{3xy^2}$$

□

Example . Find $\frac{dy}{dx}$ given that $x^2 + 2xy - 2y^2 + x = 2$ at the point $(-4, 1)$

Solution: differentiating both sides with respect to x , we have

$$\begin{aligned}\frac{d}{dx}(x^2) + \frac{d}{dx}(2xy) - \frac{d}{dx}(2y^2) + \frac{dx}{dx} &= \frac{d}{dx}(2) \\ \Rightarrow 2x + 2(y + xy') - 4yy' + 1 &= 0 \\ &= 2x + 2y + 2xy' - 4yy' + 1 = 0 \Rightarrow 2xy' - 4yy' = \\ &= -(2x + 2y + 1) \\ &= y'(2x - 4y) = -(2x + 2y + 1) \text{ or } y' = \frac{-(2x+2y+1)}{(2x-4y)} \\ \text{at the point } (-4, 1) \quad y' &= \frac{-(-8+2+1)}{(-8-4)} = -5/12\end{aligned}$$

□

Example . Find $\frac{dy}{dx}$ given that $x^2 + y^2 - 6xy + 3x - 2y + 5 = 0$





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Solution: differentiating both sides with respect to x , we have

$$\Rightarrow 2x + 2yy' - 6(y + xy') + 3 - 2y' = 0$$

$$= 2x + 2yy' - 6y - 6xy' + 3 - 2y' = 0 \Rightarrow y'(2y - 6x - 2) = 6y - 2x - y$$

$$\text{or } y' = \frac{6y - 2x - y}{(2y - 6x - 2)}$$

□

EXERCISE 1.  Find the shape of the tangent to the curve at point $(1, 2)$ given $x^2 + xy + y^2 = 7$

More Revision Questions

Find $\frac{dy}{dx}$ if;

(1) $4x^2 + 2y^2 = 9$

(2) $x^2 + 2xy + y^2 = 3$

(3) $x^2 - 3xy + y^2 - 2y + 4x = 0$

(4) $3x^2 - 4xy = 7$



$$(5) \quad x^2 + y^2 - 6xy - 2y + 3x + 5 = 0 \quad (6)$$
$$3x^2 + 5xy + 4y^2 - 4y = 0 \text{ at } (0, 1)$$

5.0.1. PARAMETRIC EQUATIONS

Consider $x = f(t)$ and $y = g(t)$ then x and y are both functions of t .

These equations are parametric equations for x and y and the variable t is called a parameter.

Example of parametric equations are $x = 2t$, $y = t^2 + 1$ and $x = \cos^3 \theta$, $y = \sin^3 \theta$

Derivative of parametric equations $x = f(t)$ and $y = g(t)$

$$\frac{dx}{dt} = f'(t) \text{ and } \frac{dy}{dt} = g'(t), \quad \frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx}$$

Example . Find the derivative $\frac{dy}{dx}$ of $x = 2t$, $y = t^2 - 1$



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Solution: $\frac{dx}{dt} = 2$ and $\frac{dy}{dt} = 2t$,

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = (2t) \frac{1}{2} = t$$

□

Example  Differentiate $x = t^3 + t^2$, $y = t^2 + t$

Solution: $\frac{dx}{dt} = 3t^2 + 2t$ and $\frac{dy}{dt} = 2t + 1$,

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = (2t + 1) \frac{1}{(3t^2 + 2t)} = \frac{2t + 1}{3t^2 + 2t}$$

□

EXERCISE 2.  Find $\frac{dy}{dx}$ if $x = \cos t$, $y = \sin t$

Example  Find the gradient of the curve $x = \frac{t}{1+t}$, $y = \frac{t^3}{1+t}$ at the point $(\frac{1}{2}, \frac{1}{2})$

Solution: $\frac{dx}{dt} = \frac{(1+t)1-t}{(1+t)^2} = \frac{1}{(1+t)^2}$

$$\frac{dy}{dt} = \frac{(1+t)3t^2 - t^2}{(1+t)^2} = \frac{3t^2 + 3t^3 - t^2}{(1+t)^2} = \frac{3t^3 + 2t^2}{(1+t)^2}$$





$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \frac{3t^3 + 2t^2}{(1+t)^2} \bigg/ \frac{1}{(1+t)^2} = 3t^3 + 2t^2$$

when $x = 1/2$ and $y = 1/2$ then $\frac{t}{1+t} = 1/2$ and $\frac{t^3}{1+t} = 1/2$

$$\Rightarrow 2t = 1 + t \Rightarrow t = 1 \text{ and } 2t^3 = 1 + t \Rightarrow t = 1$$

when $t = 1$, $\frac{dy}{dx} = 3t^3 + 2t^2 = 3 + 2 = 5$

5.0.2. More Revision Questions

Find $\frac{dy}{dx}$ in terms of t

1. $x = t^3 + t^2$ $y = t^2 + t$

2. $x = at^2$ $y = 2at$

3. $x = (1 + t)^2$ $y = t^2 - 1$

4. $x = \cos^2 2t$ $y = \sin^2 2t$

5. $x = t^2 - \pi/2$ $y = \sin t^2$



5.1. Derivatives of trigonometric functions

Recall $\cos^2 x + \sin^2 x = 1$, $\frac{1}{\cos x} = \sec x$ and $\frac{1}{\sin x} = \csc x$
 $\cot^2 x + 1 = \csc^2 x$ $1 + \tan^2 x = \sec^2 x$ $\frac{\sin x}{\cos x} = \tan x$, $\frac{\cos x}{\sin x} = \cot x$

Example Derivative of $\tan x$

Solution: $y = \tan x$ $\tan x = \frac{\sin x}{\cos x} = \frac{u}{v}$ $\frac{dy}{dx} =$
 $\frac{\cos x \cos x - \sin x(-\sin x)}{(\cos x)^2} = \frac{\cos^2 x + \sin^2 x}{(\cos x)^2} = \frac{1}{(\cos x)^2} = \sec^2 x$ □

Example Derivative of $\cot x$

Solution: $y = \cot x$ $\cot x = \frac{\cos x}{\sin x} = \frac{u}{v}$
 $\frac{dy}{dx} = \frac{\sin x(-\sin x) - \cos x(\cos x)}{(\sin x)^2} = \frac{-(\sin^2 x + \cos^2 x)}{(\sin x)^2} = \frac{-1}{(\sin x)^2} = -\csc^2 x$



5.2. Exponential function

$$y = e^x$$

$$\text{But } e^x = 1 + x + x^2/2! + x^3/3! + x^4/4! + \cdots + x^h/h! + \cdots$$

$$\frac{dy}{dx} = \frac{d}{dx} (e^x) = 1 + x + x^2/2! + x^3/3! + x^4/4! + \cdots + x^{h-1}/(h-1)! + \cdots$$

If $y = e^{ax}$ where a is a constant

$$\text{Let } u = ax \Rightarrow \frac{du}{dx} = a$$

$$y = e^u \Rightarrow \frac{dy}{du} = e^u$$

$$\frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx} = e^u a = ae^{ax}$$

5.3. Logarithmic function

$$f(x) = \ln x$$

$$\text{If } y = \ln x \text{ then } x = e^y, \text{ and } \frac{dx}{dy} = e^y \text{ and } \frac{dy}{dx} = \frac{1}{e^y} = \frac{1}{x}$$



$$\therefore \frac{dy}{dx} = \frac{d \ln x}{dx} = \frac{1}{x}$$

Example . Write down the derivative of $\ln(2x^3)$

Solution: Let $y = \ln(2x^3)$

Then $y = \ln 2 + \ln x^3$ and $y = \ln 2 + 3 \ln x$

$$\frac{dy}{dx} = \frac{3}{x} \text{ or } \frac{dy}{dx} = \frac{6x}{2x^2} = \frac{3}{x}$$

Example . Write down the derivative of $\ln\left(\frac{1}{\sqrt{x}}\right)$

Solution: Let $y = \ln\left(\frac{1}{\sqrt{x}}\right)$

Then $y = \ln 1 - \ln \sqrt{x}$ or $y = \ln 1 - \frac{1}{2} \ln x$ and $\frac{dy}{dx} = 0 - \frac{1}{2} \frac{1}{x}$

or $\frac{dy}{dx} = \frac{-1}{2x}$



More Revision Questions

Write down the derivatives of

- a) $2 \ln x$ b) $\ln x^5$ c) $\ln(3x^2)$ d) $\ln(x^{-1/2})$
- e) $\ln(3/\sqrt{x})$ f) $\ln(3x/\sqrt{x})$ g) $\ln \sqrt{(2/x^5)}$ h) $\ln \sqrt{\frac{1}{3}x^{-\frac{1}{2}}}$



Solutions to Exercises

Exercise 1. $2x + y + x \frac{dy}{dx} + 2y \frac{dy}{dx} = 0, \quad \frac{dy}{dx} (x + 2y) = -2x - y$

OR

$$\frac{dy}{dx} = \frac{-2x-y}{x+2y} = \frac{-2-2}{1+4} = \frac{-4}{5}$$

Exercise 1



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Exercise 2. $\frac{dx}{dt} = -\sin t$ and $\frac{dy}{dt} = \cos t$,

$$\frac{dy}{dx} = \frac{dy}{dt} \frac{dt}{dx} = \cos t \frac{1}{-\sin t} = -\frac{\cos t}{\sin t}$$

Exercise 2